

ECN 591: Market Design Problem Set 2

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Part 1: Market Setup

Student Preferences

Table 1: Student Preference Profile

Student	1st Choice	2nd Choice	3rd Choice
i1	s3	s3	s3
i10	s3	s3	s3
i11	s3	s3	s3
i12	s2	s2	s2
i13	s2	s2	s2
i14	s3	s3	s3
i15	s1	s1	s1
i16	s3	s3	s3
i17	s1	s1	s1
i18	s1	s1	s1
i2	s3	s3	s3
i3	s2	s2	s2
i4	s3	s3	s3
i5	s1	s1	s1
i6	s3	s3	s3
i7	s3	s3	s3
i8	s1	s1	s1
i9	s1	s1	s1

School Priorities

Table 2: School Priority Orderings

School	Priority 1	Priority 2	Priority 3	Priority 4	Priority 5
s1	i5	i10	i8	i11	i12
s2	i15	i10	i18	i1	i3
s3	i2	i7	i16	i3	i5

Part 2: Matching Mechanisms

Final Matching Outcomes

Table 3: Final Assignment Under Each Mechanism

Student	DA	IA	TTC
i1	s1	s2	s3
i10	s3	s3	s3
i11	s3	s1	s3
i12	s1	s2	s2
i13	s3	s2	s2
i14	s1	s3	s3
i15	s2	s1	s1
i16	s3	s3	s3
i17	s2	s1	s1
i18	s2	s1	s1
i2	s1	s3	s3
i3	s2	s2	s2
i4	s1	s2	s3
i5	s3	s1	s1
i6	s2	s3	s3
i7	s2	s3	s3
i8	s3	s1	s1
i9	s1	s1	s1

Part 3: Efficiency Analysis

Rank of Assigned School

Table 4: Rank of Assigned School by Mechanism

Student	DA Rank	IA Rank	TTC Rank
i1	2	2	1
i10	1	4	1
i11	2	2	1
i12	1	1	1
i13	1	1	1
i14	1	1	1
i15	1	1	1
i16	1	1	1
i17	1	1	1
i18	1	1	1
i2	1	1	1
i3	1	1	1
i4	3	3	1
i5	1	1	1
i6	1	1	1
i7	1	1	1
i8	1	1	1
i9	2	3	1

Average Rank Across 1000 Simulations

Table 5: Average Student Rank

Mechanism	Average Rank
Deferred Acceptance (DA)	1.2127
Immediate Acceptance (IA)	1.2694
Top Trading Cycles (TTC)	4.0000

Interpretation

Looking at Table 5, Deferred Acceptance (DA) has the lowest ranked average of 1.213 out of the three matching mechanisms with Immediate Acceptance (IA) ranked as the second lowest with 1.270 and Top Trading Cycles (TTC) ranked as the highest average with 4.000. An average DA of 1.213 shows that students do not receive optimal assignments under this mechanism compared to IA which is slightly better and TTC which has a more optimal outcome.